Worksheet No. 10

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Subject Name: Technical Training **Subject Code:** 25CAP-652

1.Aim/Overview of the practical:

To write a transaction that performs multiple database operations.

2. Objectives:

To write a transaction that performs multiple database operations.

3. Input/Apparatus Used:

- PostgreSQL
- pgAdmin

4.Procedure/Algorithm/Code:

a. Table Creation

```
CREATE TABLE Payroll (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   salary DECIMAL (10,2) CHECK(SALARY>0)
);
```

```
CREATE TABLE

Query returned successfully in 116 msec.
```

b. Inserting records into the table

```
INSERT INTO Payroll VALUES
(1, 'Amit', 30000),
(2, 'Neha', 40000),
(3, 'Ravi', 50000);
```

c.**View the records**

```
SELECT * FROM PAYROLL
```

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	1	Amit	30000.00
2	2	Neha	40000.00
3	3	Ravi	50000.00

d. Begin the transaction

BEGIN;

```
BEGIN

Query returned successfully in 101 msec.
```

e. First Update

UPDATE Payroll
SET salary = salary + 5000
WHERE emp_id = 1;

```
UPDATE 1

Query returned successfully in 64 msec.
```

SELECT * FROM Payroll;

	emp_id [PK] integer	emp_name character varying (50)	salary numeric (10,2)
1	2	Neha	40000.00
2	3	Ravi	50000.00
3	1	Amit	35000.00

f. Savepoint

SAVEPOINT sp1;

-- Update 2
UPDATE Payroll
SET salary = salary + 7000
WHERE emp_id = 2;

-- Error simulation
UPDATE Payroll
SET salary = -1000
WHERE emp_id = 3;

-- Rollback to savepoint
ROLLBACK TO sp1;

g.Commit valid changes
COMMIT;

```
COMMIT

Query returned successfully in 287 msec.
```

5. Learning Outcomes:

1. Using TCL commands COMMIT, ROLLBACK and SAVEPOINT in PostgreSQL.
2. Transactions in DBMS.